Inflation Dynamics - Computational Notebook

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Abstract

This book presents computation details of estimating inflation dynamics.

This notebook presents computation details of estimating inflation dynamics.

So far we have covered estimation of

1. GARCH
2. EGARCH
3. GJR-GARCH
4. SPARCH
5. Regime-switching GARCH
6. BEGE GARCH

State variable definitions:

1. π_t: inflation.
2. $\mathcal{F}_{t-1}$: filtration at time $t-1$.
3. SPF_{t-1}: professional forecast on π_t inflation measured by $\mathcal{F}_{t-1}$.
4. $\hat{\pi}_t$: expected inflation $E[\pi_t \mid \mathcal{F}_{t-1}]$.
5. u_t: inflation innovation (residual), defined as $u_t := \pi_t - \hat{\pi}_t$.
6. u_t^+ : postive residual, defined as $u_t^+ := u_t \cdot \mathbf{1}(u_t > 0)$
7. u_t^- : negative residual, defined as $u_t^- := u_t \cdot \mathbf{1}(u_t \leq 0)$

Computation Notebook

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1. Inflation Summary Statistics

I start with simple statistical summary by plotting inflation and professional forecast over time.

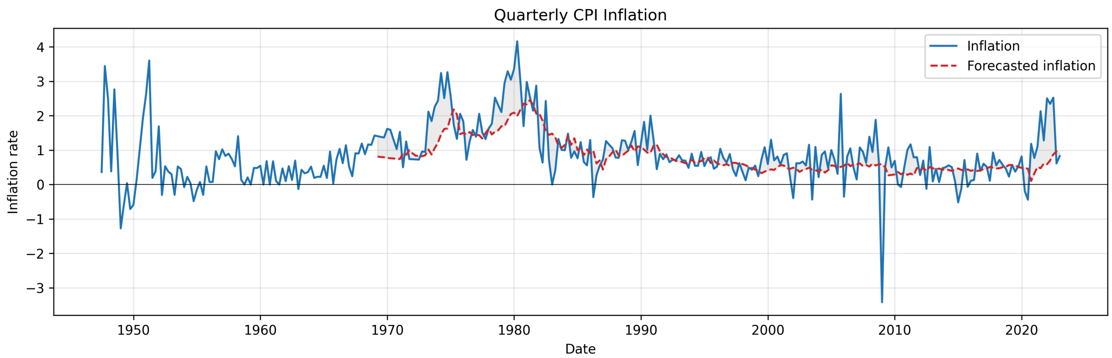

Figure 1: Quarterly inflation π_t and profesional forcast SPF_{t-1}.

Notice that inflation was recorded from 1947 but profesional forecast started from 1969.

	Inflation	Forecasted inflation
Date Start	1947Q2	1969Q1
Date End	2022Q4	2022Q4
#Observations	303	216

Table 1: Inflation statistics

1.a. Effective Sample

To make log-likelihood values comparable across specifications, I use an effective sample for all mean models and all volatility models.

In ARX models, lagged regressors (for example, SPF_{t-1}) are not observed at the very beginning of the raw sample. Therefore, I start the estimation sample at the first date where both current forecast inflation (SPF_t), and one-lag forecast inflation (SPF_{t-1}) are available. For the quarterly data, forecast inflation (SPF_t) is available from **1969Q1** to **2022Q4** (216 observations), requiring one lag (SPF_{t-1}) shifts the usable start to **1969Q2**. So the trimmed estimation window is **1969Q2–2022Q4**, with **215 observations**. This same trimmed sample is then used for all mean specifications, so the residual series has the same length across models.

For GARCH-type recursion, the first few conditional variances are unobserved (for example, σ_0^2, σ_{-1}^2, u_0^2, u_{-1}^2). I initialize these pre-sample variances using the model-implied unconditional variance, for example in GARCH model:

$$\bar{\sigma}^2 = \frac{\omega}{1 - \sum_i \alpha_i - \sum_j \beta_j}, \tag{1}$$

and set

$$u_0^2 = u_{-1}^2 = \cdots = \sigma_0^2 = \sigma_{-1}^2 = \cdots = \bar{\sigma}^2. \tag{2}$$

1.b. Effective Sample Summary Statistics

I report the summary statistics for trimmed effect sample. Skewness and Kurtosis are adjusted by sample size.

	Inflation	SPF	SPF_shock
Date Start	1969Q2	1969Q2	1969Q2
Date End	2022Q4	2022Q4	2022Q4
#Observations	215	215	215
Mean	0.9918	0.8320	0.1598
Median	0.7937	0.6296	0.1101
Std	0.8580	0.4969	0.6642
Min	−3.4170	0.1048	−4.0117
Max	4.1612	2.4566	2.1729
P5	−0.0681	0.3482	−0.7745
P25	0.5328	0.4748	−0.1669
P75	1.2913	1.0150	0.4617
P95	2.5883	2.0030	1.2581
Skewness	0.3550	1.3633	−0.7131
Kurtosis	3.8995	1.2053	7.3006
AC(1)	0.5981	0.9717	0.2764
AC(2)	0.5353	0.9473	0.1322
AC(4)	0.4708	0.8929	0.0881
AC(12)	0.2918	0.6949	0.0011

Table 2: Inflation statistics

2. Mean Process Specification

Four types of mean processes are of interest:

- **Constant**: $\pi_{t+1} = SPF_t + \mu_{t+1}$
- **ARX(1,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,2)**[1]: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + \mu_{t+1}$

where μ_{t+1} is the residual term.

2.a. *OLS Results*

Estimation results of four mean models with HAC standard errors in parentheses below each estimate. (The log-likelihood is computed under the Gaussian assumption.)

Constant:

$$\hat{\pi}_{t+1} = SPF_t \tag{3}$$

No estimated coefficients. Residuals $\mu_{t+1} = \pi_{t+1} - SPF_t$.

- Log-likelihood: −**222.664**
- AIC: **447.329**, BIC: **450.700**
- Residual skewness: −**0.713**, excess kurtosis: **7.301**

ARX(1,1):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0824} \; \underset{(0.112)}{+0.3005\,\pi_t} \; \underset{(0.167)}{+0.7337\,SPF_t} \tag{4}$$

- Log-likelihood: −**207.381**
- AIC: **420.762**, BIC: **430.874**
- Residual skewness: −**1.047**, excess kurtosis: **8.210**

ARX(2,1):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0897} \; \underset{(0.098)}{+0.2892\,\pi_t} \; \underset{(0.126)}{+0.0834\,\pi_{t-1}} \; \underset{(0.251)}{+0.6385\,SPF_t} \tag{5}$$

- Log-likelihood: −**206.810**
- AIC: **421.619**, BIC: **435.102**
- Residual skewness: −**1.191**, excess kurtosis: **9.050**

ARX(2,2):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0856} \; \underset{(0.106)}{+0.2992\,\pi_t} \; \underset{(0.125)}{+0.0914\,\pi_{t-1}} \; \underset{(0.433)}{+0.4084\,SPF_t} \; \underset{(0.297)}{+0.2136\,SPF_{t-1}} \tag{6}$$

- Log-likelihood: −**206.660**
- AIC: **423.319**, BIC: **440.172**
- Residual skewness: −**1.212**, excess kurtosis: **9.129**

[1] In the GARCH family estimation exercise, we found that ARX(2,2) is always dominated by other three specifications. Thus, we stop estimating ARX(2,2) in BEGE estimation to saving computational resources.

3. GARCH Family

This section presents three GARCH-type volatility models — GARCH, GJR-GARCH, and EGARCH — each combined with one of three conditional distributions for the innovations: normal, Student's t, and a finite mixture of normals.

Let $\{u_t\}_{t=1}^{T}$ denote the residual (innovation) process, and let $\mathcal{F}_{t-1}$ denote the information set available at time $t-1$. We assume

$$u_t = \sigma_t \, z_t, \qquad z_t \texttt{ mid } \mathcal{F}_{t-1} \overset{\text{i. i. d.}}{\sim} D(0,1), \tag{7}$$

where $\sigma_t^2 = \operatorname{Var}(u_t \texttt{ mid } \mathcal{F}_{t-1})$ is the conditional variance and $D(0,1)$ is a standardized distribution (normal, Student's t, or mixture of normals) with zero mean and unit variance. Throughout, $z_t = u_t / \sigma_t$ denotes the standardized residual and θ denotes the full parameter vector (volatility-recursion parameters together with any distributional parameters).

3.a. *Volatility Recursion Specifications*

GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \, u_{t-i}^2 + \sum_{k=1}^{q} \beta_k \, \sigma_{t-k}^2, \tag{8}$$

with $\omega > 0$, $\alpha_i \geq 0$, $\beta_k \geq 0$, and $\sum_i \alpha_i + \sum_k \beta_k < 1$ for covariance stationarity.

GJR-GARCH(p, o, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \, u_{t-i}^2 + \sum_{j=1}^{o} \gamma_j \, u_{t-j}^2 \, \mathbf{1}\{u_{t-j} < 0\} + \sum_{k=1}^{q} \beta_k \, \sigma_{t-k}^2, \tag{9}$$

where $\mathbf{1}\{\cdot\}$ is the indicator function. The term γ_j captures the leverage effect: when $\gamma_j > 0$, negative shocks raise future volatility more than positive shocks of equal magnitude.

EGARCH(p, o, q):

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \left(| \, z_{t-i} \, | - \mathbb{E} \, | \, z_{t-i} \, | \right) + \sum_{j=1}^{o} \gamma_j \, z_{t-j} + \sum_{k=1}^{q} \beta_k \ln \sigma_{t-k}^2. \tag{10}$$

Under the standard normal distribution, $\mathbb{E} \, | \, z_t \, | = \sqrt{2/\pi}$; for other distributions the corresponding constant should be used. Modeling $\ln \sigma_t^2$ guarantees positivity of σ_t^2 without parameter restrictions on $(\alpha_i, \gamma_j, \beta_k)$.

3.b. *Log-Likelihood Specifications*

For each candidate distribution we give (i) the standardized density of z_t, (ii) the conditional density of $u_t \texttt{ mid } \mathcal{F}_{t-1}$ obtained via the change of variables $u_t = \sigma_t z_t$ (which contributes a Jacobian factor $1/\sigma_t$), (iii) the per-observation log-likelihood $\ell_t(\theta)$, and (iv) the sample log-likelihood

$$\ln L(\theta) = \sum_{t=1}^{T} \ell_t(\theta). \tag{11}$$

3.b.i. *Normal distribution:*

The standardized innovation satisfies $z_t \texttt{ mid } \mathcal{F}_{t-1} \sim \mathcal{N}(0,1)$ with density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\}. \tag{12}$$

By the change of variables $u_t = \sigma_t z_t$, the conditional density of u_t is

$$f(u_t \ \mathtt{mid}\ \mathcal{F}_{t-1};\theta) = \frac{1}{\sigma_t}\phi\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sqrt{2\pi\,\sigma_t^2}}\exp\left\{-\frac{u_t^2}{2\,\sigma_t^2}\right\}. \tag{13}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\frac{1}{2}\left(\ln 2\pi + \ln\sigma_t^2 + \frac{u_t^2}{\sigma_t^2}\right), \tag{14}$$

and the sample log-likelihood is

$$\ln L(\theta) = -\frac{1}{2}\sum_{t=1}^{T}\left(\ln 2\pi + \ln\sigma_t^2 + \frac{u_t^2}{\sigma_t^2}\right). \tag{15}$$

3.b.ii. *Student's t distribution:*

Let $\nu > 2$ denote the degrees-of-freedom parameter. The standardized Student's t density (rescaled to unit variance) is

$$f_\nu(z) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)}}\left(1+\frac{z^2}{\nu-2}\right)^{-\frac{\nu+1}{2}}, \tag{16}$$

where $\Gamma(\cdot)$ is the gamma function. The restriction $\nu > 2$ ensures finite variance. The conditional density of u_t is

$$f(u_t \ \mathtt{mid}\ \mathcal{F}_{t-1};\theta) = \frac{1}{\sigma_t} f_\nu\left(\frac{u_t}{\sigma_t}\right) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)\,\sigma_t^2}}\left(1+\frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right)^{-\frac{\nu+1}{2}}. \tag{17}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = \ln\Gamma\left(\frac{\nu+1}{2}\right) - \ln\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\ln\left(\pi(\nu-2)\sigma_t^2\right) - \frac{\nu+1}{2}\ln\left(1+\frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right), \tag{18}$$

and the sample log-likelihood is

$$\ln L(\theta) = T\left[\ln\Gamma\left(\frac{\nu+1}{2}\right) - \ln\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\ln\left(\pi(\nu-2)\right)\right] - \frac{1}{2}\sum_{t=1}^{T}\ln\sigma_t^2 - \frac{\nu+1}{2}\sum_{t=1}^{T}\ln\left(1+\frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right) \tag{19}$$

3.b.iii. *Mixture of two normals:*

The standardized innovation follows a two-component Gaussian mixture with parameters (p_1, μ_1, σ_1^2):

$$f_{\text{mix}}(z) = p_1\,\phi(z;\mu_1,\sigma_1^2) + p_2\,\phi(z;\mu_2,\sigma_2^2), \qquad \phi(z;\mu,s^2) = \frac{1}{\sqrt{2\pi s^2}}\exp\left\{-\frac{(z-\mu)^2}{2s^2}\right\}. \tag{20}$$

To enforce $\mathbb{E}[z_t] = 0$ and $\mathrm{Var}(z_t) = 1$, the remaining parameters are determined by

$$p_2 = 1 - p_1, \qquad \mu_2 = -\frac{p_1\mu_1}{p_2}, \qquad \sigma_2^2 = \frac{1 - p_1(\sigma_1^2 + \mu_1^2) - p_2\mu_2^2}{p_2}. \tag{21}$$

This leaves three free parameters: (p_1, μ_1, σ_1^2).

Hierarchical representation. The mixture admits the latent-variable form $S \sim \text{Bernoulli}(p_1)$, $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, with $z = X_1$ if $S = 1$ and $z = X_2$ otherwise. As a consequence, moments and the CDF decompose as

$$\mathbb{E}[z^n] = p_1\,\mathbb{E}[X_1^n] + p_2\,\mathbb{E}[X_2^n], \qquad F_{\text{mix}}(a) = p_1\Phi\left(\frac{a-\mu_1}{\sigma_1}\right) + p_2\Phi\left(\frac{a-\mu_2}{\sigma_2}\right), \tag{22}$$

where Φ is the standard normal CDF.

The conditional density of u_t is

$$f(u_t \ \texttt{mid}\ \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t} f_{\text{mix}}\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sigma_t}\left[p_1\,\phi(z_t; \mu_1, \sigma_1^2) + p_2\,\phi(z_t; \mu_2, \sigma_2^2)\right], \qquad z_t = \frac{u_t}{\sigma_t}. \tag{23}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\ln\sigma_t + \ln\left[p_1\,\phi(z_t; \mu_1, \sigma_1^2) + p_2\,\phi(z_t; \mu_2, \sigma_2^2)\right], \tag{24}$$

and the sample log-likelihood is

$$\ln L(\theta) = -\sum_{t=1}^{T}\ln\sigma_t + \sum_{t=1}^{T}\ln\left[p_1\,\phi(z_t; \mu_1, \sigma_1^2) + p_2\,\phi(z_t; \mu_2, \sigma_2^2)\right]. \tag{25}$$

4. GARCH Family

This section presents three GARCH-type volatility models — GARCH, GJR-GARCH, and EGARCH — each combined with one of three conditional distributions for the innovations: normal, Student's t, and a finite mixture of normals.

Let $\{u_t\}_{t=1}^{T}$ denote the residual (innovation) process, and let $\mathcal{F}_{t-1}$ denote the information set available at time $t-1$. We assume

$$u_t = \sigma_t\, z_t, \qquad z_t \ \texttt{mid}\ \mathcal{F}_{t-1} \overset{\text{i. i. d.}}{\sim} D(0, 1), \tag{26}$$

where $\sigma_t^2 = \operatorname{Var}(u_t \ \texttt{mid}\ \mathcal{F}_{t-1})$ is the conditional variance and $D(0, 1)$ is a standardized distribution (normal, Student's t, or mixture of normals) with zero mean and unit variance.

4.a. Volatility Recursion Specification

GARCH(p,q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p}\alpha_i u_{t-i}^2 + \sum_{k=1}^{q}\beta_k\sigma_{t-k}^2 \tag{27}$$

GJR-GARCH(p,q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p}\alpha_i \mid u_{t-i}^{+} \mid^2 + \sum_{j=1}^{p}\gamma_j \mid u_{t-j}^{-} \mid^2 + \sum_{k=1}^{q}\beta_k\sigma_{t-k}^2 \tag{28}$$

EGARCH(p,q):

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \left(\mid e_{t-i} \mid - \sqrt{2/\pi} \right) + \sum_{j=1}^{o} \gamma_j e_{t-j} + \sum_{k=1}^{q} \beta_k \ln \sigma_{t-k}^2 \tag{29}$$

where $e_{t-j} = u_{t-j}/\sigma_{t-j}$ is the standardized residual.

4.b. Log Likelihood Specification

4.b.i. *Normal distribution:*

The residual u_t follows normal distribution with mean zero and volitility σ_t conditional on $\mathcal{F}_{t-1}$,

$$u_t \mid \mathcal{F}_{t-1} \sim N(0, \sigma_t^2). \tag{30}$$

The probability density function of u_t is

$$f(u_t \mid \mathcal{F}_{t-1}) = \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left\{ -\frac{u_t^2}{2\sigma_t^2} \right\}. \tag{31}$$

And the log likelihood given residual u_t is

$$\log l(\theta \mid u_t) = -\frac{1}{2} \left(\log 2\pi + \log \sigma_t^2 + \frac{u_t^2}{\sigma_t^2} \right) \tag{32}$$

The sample log likelihood given residuals $\{u_1, u_2, ..., u_T\}$ is

$$\ln L = -\frac{1}{2} \sum_{t=1}^{T} \left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2} \right) \tag{33}$$

4.b.ii. *Student t distribution:*

One parameter : ν

Log Likelihood Function of one residual ε_t with conditional standard error σ_t :

$$\ln \Gamma\left(\frac{\nu+1}{2}\right) - \ln \Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2} \ln(\pi(\nu-2)\sigma_t^2) - \frac{\nu+1}{2} \ln(1 + \varepsilon_t^2/(\sigma_t^2(\nu-2))) \tag{34}$$

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where $\Gamma$ is the gamma function.
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4.b.iii. *Gaussian Mixture distribution:*

The mixture of 2 normal distribution given parameters p_1,μ_1,σ_1^2 is

$$f(z_t) = p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} exp\left\{ -\frac{(z_t - \mu_1)^2}{2\sigma_1^2} \right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} exp\left\{ -\frac{(z_t - \mu_2)^2}{2\sigma_2^2} \right\} \tag{35}$$

where $p_2 = 1 - p_1$,$\mu_2 = \frac{-p_1\mu_1}{p_2}$, and $\sigma_2^2 = \frac{1 - p_2\mu_2^2 - p_1(\sigma_1^2 + \mu_1^2)}{p_2}$

N th order moment can be calculated as:

$$E[z^n] = \int z^n \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} exp\left\{ -\frac{(z-\mu_1)^2}{2\sigma_1^2} \right\} + (1-p_1) \frac{1}{\sqrt{2\pi\sigma_2^2}} exp\left\{ -\frac{(z-\mu_2)^2}{2\sigma_2^2} \right\} \right] dz \tag{36}$$

$$E[z^n] = p_1 E[x_1^n \mid x_1 \sim N(\mu_1, \sigma_1^2)] + p_2 E[x_2^n \mid x_2 \sim N(\mu_2, \sigma_2^2)] \tag{37}$$

Similary, CDF can be calculated analytically by

$$\Phi(a) = \int_{-\inf}^{a} z \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} exp\left\{ -\frac{(z-\mu_1)^2}{2\sigma_1^2} \right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} exp\left\{ -\frac{(z-\mu_2)^2}{2\sigma_2^2} \right\} \right] dz \tag{38}$$

$$\Phi(a) = p_1 \Phi_{x_1}(a) + p_2 \Phi_{x_2}(a) \tag{39}$$

We can think mixture of two normal distributions generated by three random variable Z,X_1,X_2. Z follows Bernoulli(P_1), $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$. In order to generate a random sample point, we can first use Z to decide which normal distributions to use, and then randomly pick a point in that normal distribution.

Log likelihood function is

$$L(u_t) = \sum_{t=1}^{T} \ln \frac{1}{\sigma_t} \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} exp\left\{ -\frac{(z_t-\mu_1)^2}{2\sigma_1^2} \right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} exp\left\{ -\frac{(z_t-\mu_2)^2}{2\sigma_2^2} \right\} \right] \tag{40}$$

, where $z_t = \frac{u_t}{\sigma_t}$

4.c. Estimation Summary

4.d. Best Estimation Illustrations

Table 2: Model selection by AIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	(1,1)	(2,1)	359.4710	(1,1)	(1,1)	368.2429
t	(1,1)	(2,1)	343.1431	(2,1)	(1,1)	344.6254
Skew t	(1,1)	(2,1)	345.1312	(2,1)	(1,1)	346.5791
GED	(1,1)	(2,1)	345.1925	(1,1)	(1,1)	347.6934
Mix of Normal	(1,1)	(2,1)	345.8222	(2,1)	(2,1)	343.5023

Table 3: Model selection by BIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	FC	(2,1)	381.4280	FC	(1,1)	383.4754
t	FC	(1,1)	367.8350	FC	(1,1)	367.1732
Skew t	FC	(1,1)	370.4242	FC	(1,1)	369.6649
GED	FC	(1,1)	371.8489	FC	(1,1)	370.9644
Mix of Normal	FC	(1,1)	373.7106	FC	(1,1)	371.0689

5. BEGE Density

BEGE density function $f_{BEGE}(\mu \mid p, n, \sigma_p, \sigma_n)$ is the function that calculates the density of the observation μ given the parameters $\{p, n, \sigma_p, \sigma_n\}$.

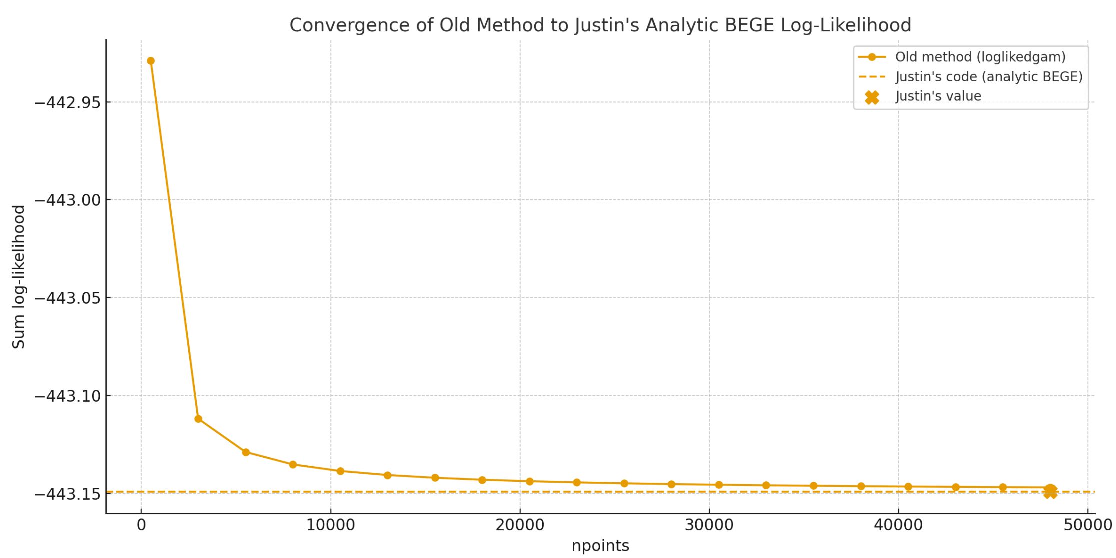
To compare Justin's analytic BEGE density code with the old numerical one, I conducted an experiment using synthetic data. I generated 200 independent observations from a standard normal distribution and fixed the BEGE parameters to

$$p = 5, \qquad n = 7, \qquad \sigma_p = 0.8, \qquad \sigma_n = 1.2. \tag{41}$$

I computed the sum of log-likelihood using two approaches:

1. The *old BEGE function*, which approximates the density via numerical integration on a uniform grid, and whose accuracy depends strongly on the number of grid points; and
2. *Justin's analytic BEGE function*.

The old numerical method was evaluated over a wide range of grid resolutions (up to $50,000$ points), and the resulting log-likelihood sums were plotted alongside the benchmark value computed from Justin's code.



Findings:

- The old numerical likelihood converges toward Justin's analytic likelihood as the number of grid points increases, but the convergence is slow and requires very fine grids.
- For small or moderate grid sizes, the numerical integration *overestimates* the log-likelihood. This explains why, for a given set of parameters, Justin's implementation produces a lower log-likelihood than the old code.
- Justin's analytic BEGE formula provides a more accurate and more computationally efficient evaluation.

6. BEGE GARCH Family

To start the random search of initial mean parameters, I draw uniform samples from $(\mu - 2\sigma,\ \mu + 2\sigma)$ where μ and σ are mean and standard deviation from OLS regression. I also set the AR coefficient bound to avoid the AR process to explode. We have four types of mean processes.

Table 1: Parameter Bounds for Mean Process Specifications

Model	c	ρ_1	ρ_2	ϕ_1	ϕ_2
Constant	–	–	–	–	–
ARX(1,1)	$(\min \pi_t,\ \ \max \pi_t)$	$(-0.999,\ 0.999)$	–	$(-10,\ 10)$	–
ARX(2,1)	$(\min \pi_t,\ \ \max \pi_t)$	$(-1.999,\ 1.999)$	$(-0.999,\ 0.999)$	$(-10,\ 10)$	–
ARX(2,2)	$(\min \pi_t,\ \ \max \pi_t)$	$(-1.999,\ 1.999)$	$(-0.999,\ 0.999)$	$(-10,\ 10)$	$(-10,\ 10)$

6.a. Constant BEGE

The first model that I estimated is BEGE with invariant shape parameters, which is

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1} \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n \end{aligned} \tag{42}$$

where

$$\omega_p \sim \tilde{\Gamma}(\bar{p}, 1), \quad \omega_n \sim \tilde{\Gamma}(\bar{n}, 1). \tag{43}$$

$\bar{p}$ and $\bar{n}$ are unconditional shape parameters. The random search and bounds are set to be:

- $0.05 < \sigma_p, \sigma_n < 2$
- $0.1 < p, n < 10$

6.b. Symmetric BEGE

The model assumes different scaled parameters and different constants in GJR recursions for both components:

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p\, \omega_p - \sigma_n\, \omega_n, \end{aligned} \tag{44}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1). \tag{45}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho, \phi^+, \phi^-)$:

$$\begin{aligned} p_t &= p_0 + \rho\, p_{t-1} + \frac{\phi^+}{2\sigma_p^2} \left(u_{t-1}^+\right)^2 + \frac{\phi^-}{2\sigma_p^2} \left(u_{t-1}^-\right)^2, \\ n_t &= n_0 + \rho\, n_{t-1} + \frac{\phi^+}{2\sigma_n^2} \left(u_{t-1}^+\right)^2 + \frac{\phi^-}{2\sigma_n^2} \left(u_{t-1}^-\right)^2 \end{aligned} \tag{46}$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10,$
- $10^{-5} < \rho < 0.999,$
- $10^{-5} < \phi^+, \phi^- < 0.999,$

- $10^{-5} < \sigma_p, \sigma_n < 2$.

6.c. Bad and Good Symmetric GARCH Model

Here n_t, p_t is generated recursively with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p, \phi_n)$:

$$\begin{aligned} p_t &= p_0 + \rho_p\, p_{t-1} + \frac{\phi_p}{2\sigma_p^2}\,(u_{t-1})^2, \\ n_t &= n_0 + \rho_n\, n_{t-1} + \frac{\phi_n}{2\sigma_n^2}\,(u_{t-1})^2 \end{aligned} \tag{47}$$

I have set the constraints below.

- $\rho + \phi < 1$ for both good and bad shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \mathrm{Var}(\pi_t)$.
- $\max\{p_t, n_t\} < 200$

6.d. Inflation and Deflation Model

Here n_t, p_t is generated recursively with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_n^-)$:

$$\begin{aligned} p_t &= p_0 + \rho_p\, p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2}\,(u_{t-1}^+)^2, \\ n_t &= n_0 + \rho_n\, n_{t-1} + \frac{\phi_n^-}{2\sigma_n^2}\,(u_{t-1}^-)^2 \end{aligned} \tag{48}$$

According to the emails, I have set the constraints below.

- $\rho_p + \frac{\phi_p^+}{2} < 1$ and $\rho_n + \frac{\phi_n^-}{2} < 1$.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \mathrm{Var}(\pi_t)$.
- $\max\{p_t, n_t\} < 200$

6.e. Full BEGE

BEGE GARCH has mean process and higher:

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p\, \omega_p - \sigma_n\, \omega_n, \end{aligned} \tag{49}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1). \tag{50}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^-)$:

$$\begin{aligned} p_t &= p_0 + \rho_p\, p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2}\,(u_{t-1}^+)^2 + \frac{\phi_p^-}{2\sigma_p^2}\,(u_{t-1}^-)^2, \\ n_t &= n_0 + \rho_n\, n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2}\,(u_{t-1}^+)^2 + \frac{\phi_n^-}{2\sigma_n^2}\,(u_{t-1}^-)^2 \end{aligned} \tag{51}$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho_p, \rho_n < 0.999$,

- $10^{-5} < \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^- < 0.999$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

According to the emails, I have set the constraints below.

- $\rho + \frac{\phi^+}{2} + \frac{\phi^-}{2} < 1$ for both BE and GE shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \mathrm{Var}(\pi_t)$ where $\mathrm{Var}(\pi_t) = 0.75$ is the unconditional variance of inflation.
- $\max\{p_t, n_t\} < 200$.

7. Result Summary

Overall, most of the estimations with random initial values converge. I randomly sampled $100,000$ starting values for the constant BEGE model, and sampled $10,000$ for symmetric BEGE because it is more time consuming. I report the best estimation below.